CORE PURE / PROOF BY INDUCTION

Proof by Induction - Series

Worked solutions 10 questions · 73 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

Questions

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1. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^{n} r\,2^r = 2 + (n-1)2^{n+1}$$ **[6]**

Solution

We prove the result by mathematical induction.

Base case: $n = 1$. The left-hand side is

$$\sum_{r=1}^{1} r\,2^r = 1 \times 2 = 2$$

and the right-hand side is

$$2 + (1-1)2^{1+1} = 2 + 0 = 2$$

so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \geq 1$. That is, assume

$$\sum_{r=1}^{k} r\,2^r = 2 + (k-1)2^{k+1}$$

Inductive step: Starting with the left-hand side for $n = k + 1$:

$$\begin{aligned}
\sum_{r=1}^{k+1} r\,2^r &= \sum_{r=1}^{k} r\,2^r + (k+1)2^{k+1} \\
&= \big(2 + (k-1)2^{k+1}\big) + (k+1)2^{k+1} \\
&= 2 + \big((k-1) + (k+1)\big)2^{k+1} \\
&= 2 + 2k\,2^{k+1} \\
&= 2 + k\,2^{k+2} \\
&= 2 + \big((k+1) - 1\big)2^{(k+1)+1}
\end{aligned}$$

This is the right-hand side with $n = k + 1$, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \geq 1$.

2. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^{n}(2^r + 2 \times 3^r) = 2^{n+1} + 3^{n+1} - 5$$ **[6]**

Solution

We use proof by induction on n.

Base case: $n = 1$. The left-hand side is

$$\sum_{r=1}^{1}(2^r + 2 \times 3^r) = 2^1 + 2 \times 3^1 = 2 + 6 = 8$$

and the right-hand side is

$$2^{1+1} + 3^{1+1} - 5 = 2^2 + 3^2 - 5 = 4 + 9 - 5 = 8$$

so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \geq 1$. That is, assume

$$\sum_{r=1}^{k}(2^r + 2 \times 3^r) = 2^{k+1} + 3^{k+1} - 5$$

Inductive step: Starting with the left-hand side for $n = k + 1$:

$$\sum_{r=1}^{k+1}(2^r + 2 \times 3^r) = \sum_{r=1}^{k}(2^r + 2 \times 3^r) + \left(2^{k+1} + 2 \times 3^{k+1}\right)$$

Now use the inductive hypothesis:

$$\begin{aligned}\sum_{r=1}^{k+1}(2^r + 2 \times 3^r) &= \left(2^{k+1} + 3^{k+1} - 5\right) + \left(2^{k+1} + 2 \times 3^{k+1}\right) \\ &= 2^{k+1} + 2^{k+1} + 3^{k+1} + 2 \times 3^{k+1} - 5 \\ &= 2^{k+2} + 3 \times 3^{k+1} - 5 \\ &= 2^{k+2} + 3^{k+2} - 5\end{aligned}$$

This is the right-hand side with $n = k + 1$, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \geq 1$.

3. Prove by induction that for all positive integers n

$$\sum_{r=1}^{n} r(r+1)(r+2) = \frac{1}{4}n(n+1)(n+2)(n+3)$$ **[6]**

Solution

We prove the result by mathematical induction.

We want to show that for all positive integers n

$$\sum_{r=1}^{n} r(r+1)(r+2) = \frac{1}{4}n(n+1)(n+2)(n+3)$$

Base case: $n = 1$. The left-hand side is

$$\sum_{r=1}^{1} r(r+1)(r+2) = 1(2)(3) = 6$$

and the right-hand side is

$$\frac{1}{4}(1)(2)(3)(4) = 6$$

so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$\sum_{r=1}^{k} r(r+1)(r+2) = \frac{1}{4}k(k+1)(k+2)(k+3)$$

Inductive step: Starting with the left-hand side for $n = k + 1$:

$$\begin{aligned}
\sum_{r=1}^{k+1} r(r+1)(r+2) &= \sum_{r=1}^{k} r(r+1)(r+2) + (k+1)(k+2)(k+3) \\
&= \frac{1}{4}k(k+1)(k+2)(k+3) + (k+1)(k+2)(k+3) \qquad \text{(by the inductive hypothesis)} \\
&= (k+1)(k+2)(k+3)\left(\frac{k}{4}+1\right) \\
&= (k+1)(k+2)(k+3)\left(\frac{k+4}{4}\right) \\
&= \frac{1}{4}(k+1)(k+2)(k+3)(k+4)
\end{aligned}$$

This is the right-hand side with $n = k + 1$, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all positive integers n.

4. Prove by induction that for $n \in \mathbb{Z}^+$

$$\sum_{r=1}^{n} r\,3^{r-1} = \frac{1+(2n-1)3^n}{4} \qquad \textbf{[6]}$$

Solution

Let

$$S_n = \sum_{r=1}^{n} r\,3^{r-1}$$

We prove by induction that

$$S_n = \frac{1+(2n-1)3^n}{4}$$

for all $n \in \mathbb{Z}^+$.

Base case: $n = 1$. The left-hand side is

$$\sum_{r=1}^{1} r\,3^{r-1} = 1 \times 3^0 = 1$$

and the right-hand side is

$$\frac{1+(2\times 1-1)3^1}{4} = \frac{1+3}{4} = 1$$

so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$. That is, assume

$$\sum_{r=1}^{k} r\,3^{r-1} = \frac{1+(2k-1)3^k}{4}$$

Inductive step: Starting with the left-hand side for $n = k+1$:

$$\begin{aligned}\sum_{r=1}^{k+1} r\,3^{r-1} &= \sum_{r=1}^{k} r\,3^{r-1} + (k+1)3^k \\ &= \frac{1+(2k-1)3^k}{4} + (k+1)3^k\end{aligned}$$

using the inductive hypothesis.

Now write the second term over a denominator of 4:

$$\begin{aligned}\sum_{r=1}^{k+1} r\,3^{r-1} &= \frac{1+(2k-1)3^k}{4} + \frac{4(k+1)3^k}{4} \\ &= \frac{1+\big((2k-1)+4k+4\big)3^k}{4} \\ &= \frac{1+(6k+3)3^k}{4} \\ &= \frac{1+3(2k+1)3^k}{4} \\ &= \frac{1+(2k+1)3^{k+1}}{4}\end{aligned}$$

This is the right-hand side with $n = k+1$, since

$$\frac{1+(2(k+1)-1)3^{k+1}}{4} = \frac{1+(2k+1)3^{k+1}}{4}$$

so the statement is true for $n = k+1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \in \mathbb{Z}^+$.

5. **(a)** Prove by induction that, for all integers $n \geq 1$

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

[6]

(b) Hence, or otherwise, write down a factorised expression for the sum of the first $2n$ cubes

$$1^3 + 2^3 + 3^3 + \cdots + (2n)^3$$

[1]

(c) Use the formula in part (a) to write down a factorised expression for the sum of the first n even cubes

$$2^3 + 4^3 + 6^3 + \cdots + (2n)^3$$

[2]

(d) Hence, or otherwise, show that

$$1^3 + 3^3 + 5^3 + \cdots + (2n-1)^3 = an^2(bn^2 - 1)$$

where a and b are rational numbers to be determined. **[3]**

Solution

(a) We prove the result by mathematical induction.

Base case: $n = 1$. The left-hand side is

$$\sum_{r=1}^{1} r^3 = 1^3 = 1$$

and the right-hand side is

$$\frac{1}{4}(1)^2(1+1)^2 = \frac{1}{4} \times 1 \times 4 = 1$$

so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \geq 1$. That is, assume

$$\sum_{r=1}^{k} r^3 = \frac{1}{4}k^2(k+1)^2$$

Inductive step: Consider $n = k + 1$:

$$\begin{aligned}
\sum_{r=1}^{k+1} r^3 &= \sum_{r=1}^{k} r^3 + (k+1)^3 \\
&= \frac{1}{4}k^2(k+1)^2 + (k+1)^3 \\
&= (k+1)^2\left(\frac{k^2}{4} + k + 1\right) \\
&= (k+1)^2\left(\frac{k^2 + 4k + 4}{4}\right) \\
&= \frac{1}{4}(k+1)^2(k+2)^2
\end{aligned}$$

This is the right-hand side with $n = k + 1$, so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all integers $n \geq 1$.

(b) Using part (a) with n replaced by $2n$:

$$\begin{aligned} 1^3 + 2^3 + 3^3 + \cdots + (2n)^3 &= \sum_{r=1}^{2n} r^3 \\ &= \frac{1}{4}(2n)^2(2n+1)^2 \\ &= n^2(2n+1)^2 \end{aligned}$$

(c) The first n even cubes are

$$2^3 + 4^3 + 6^3 + \cdots + (2n)^3 = \sum_{r=1}^{n} (2r)^3$$

Hence

$$\begin{aligned} \sum_{r=1}^{n} (2r)^3 &= 8 \sum_{r=1}^{n} r^3 \\ &= 8\left(\frac{1}{4} n^2 (n+1)^2\right) \\ &= 2n^2(n+1)^2 \end{aligned}$$

(d) The odd cubes are found by subtracting the even cubes from the first $2n$ cubes:

$$\begin{aligned} 1^3 + 3^3 + 5^3 + \cdots + (2n-1)^3 &= \left(1^3 + 2^3 + \cdots + (2n)^3\right) - \left(2^3 + 4^3 + \cdots + (2n)^3\right) \\ &= n^2(2n+1)^2 - 2n^2(n+1)^2 \\ &= n^2\left((2n+1)^2 - 2(n+1)^2\right) \\ &= n^2\left((4n^2 + 4n + 1) - (2n^2 + 4n + 2)\right) \\ &= n^2(2n^2 - 1) \end{aligned}$$

so the sum is in the form $an^2(bn^2 - 1)$ with $a = 1$, $b = 2$.

6. **(a)** Prove by induction that, for all positive integers n

$$\sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)} = \frac{n}{2n+1}$$ **[6]**

(b) Hence, show that, for all positive integers n

$$\sum_{r=n+1}^{2n} \frac{1}{(2r-1)(2r+1)} = \frac{n}{(an+b)(cn+d)}$$

where a, b, c and d are integers to be determined. **[2]**

Solution

(a) Let

$$S_n = \sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)}$$

We prove by induction that

$$S_n = \frac{n}{2n+1}$$

for all positive integers n.

Base case: $n = 1$. The left-hand side is

$$S_1 = \frac{1}{(2\times 1-1)(2\times 1+1)} = \frac{1}{1\times 3} = \frac{1}{3}$$

and the right-hand side is

$$\frac{1}{2\times 1+1} = \frac{1}{3}$$

so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$S_k = \sum_{r=1}^{k} \frac{1}{(2r-1)(2r+1)} = \frac{k}{2k+1}$$

Inductive step: Consider S_{k+1}:

$$\begin{aligned} S_{k+1} &= \sum_{r=1}^{k+1} \frac{1}{(2r-1)(2r+1)} \\ &= \sum_{r=1}^{k} \frac{1}{(2r-1)(2r+1)} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} \\ &= S_k + \frac{1}{(2k+1)(2k+3)} \end{aligned}$$

Using the inductive hypothesis:

$$\begin{aligned} S_{k+1} &= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \\ &= \frac{k(2k+3)+1}{(2k+1)(2k+3)} \\ &= \frac{2k^2+3k+1}{(2k+1)(2k+3)} \\ &= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)} \\ &= \frac{k+1}{2k+3} \end{aligned}$$

This is the right-hand side with $n = k+1$, since

$$\frac{k+1}{2k+3} = \frac{k+1}{2(k+1)+1}$$

so the statement is true for $n = k+1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all positive integers n.

(b) Using part (a):

$$\sum_{r=n+1}^{2n} \frac{1}{(2r-1)(2r+1)} = \sum_{r=1}^{2n} \frac{1}{(2r-1)(2r+1)} - \sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)}$$

So

$$\begin{aligned} \sum_{r=n+1}^{2n} \frac{1}{(2r-1)(2r+1)} &= \frac{2n}{2(2n)+1} - \frac{n}{2n+1} \\ &= \frac{2n}{4n+1} - \frac{n}{2n+1} \\ &= \frac{2n(2n+1) - n(4n+1)}{(4n+1)(2n+1)} \\ &= \frac{4n^2+2n-4n^2-n}{(4n+1)(2n+1)} \\ &= \frac{n}{(4n+1)(2n+1)} \end{aligned}$$

so the sum is in the form $\frac{n}{(an+b)(cn+d)}$ with $a = 4$, $b = 1$, $c = 2$, $d = 1$.

7. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^{n} \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$$ **[6]**

Solution

We prove the result by induction.

Base case: $n = 1$. The left-hand side is

$$\sum_{r=1}^{1} \frac{r}{2^r} = \frac{1}{2}$$

and the right-hand side is

$$2 - \frac{1+2}{2^1} = 2 - \frac{3}{2} = \frac{1}{2}$$

so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \geq 1$. That is, assume

$$\sum_{r=1}^{k} \frac{r}{2^r} = 2 - \frac{k+2}{2^k}$$

Inductive step: We must show that

$$\sum_{r=1}^{k+1} \frac{r}{2^r} = 2 - \frac{(k+1)+2}{2^{k+1}}$$

Starting from the left-hand side:

$$\begin{aligned}
\sum_{r=1}^{k+1} \frac{r}{2^r} &= \sum_{r=1}^{k} \frac{r}{2^r} + \frac{k+1}{2^{k+1}} \\
&= \left(2 - \frac{k+2}{2^k}\right) + \frac{k+1}{2^{k+1}} \qquad \text{(by the inductive hypothesis)} \\
&= 2 - \frac{2k+4}{2^{k+1}} + \frac{k+1}{2^{k+1}} \\
&= 2 - \frac{2k+4-(k+1)}{2^{k+1}} \\
&= 2 - \frac{k+3}{2^{k+1}} \\
&= 2 - \frac{(k+1)+2}{2^{k+1}}
\end{aligned}$$

This is the right-hand side with $n = k+1$, so the statement is true for $n = k+1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \geq 1$.

8. Prove by induction that for $n \in \mathbb{Z}^+$

$$\sum_{r=1}^{n} \frac{1}{r(r+1)(r+2)} = \frac{n(n+3)}{4(n+1)(n+2)} \qquad \textbf{[6]}$$

Solution

We prove the result by induction on n.

Base case: $n = 1$. The left-hand side is

$$\sum_{r=1}^{1} \frac{1}{r(r+1)(r+2)} = \frac{1}{1 \times 2 \times 3} = \frac{1}{6}$$

and the right-hand side is

$$\frac{1(1+3)}{4(1+1)(1+2)} = \frac{4}{4 \times 2 \times 3} = \frac{1}{6}$$

so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$. That is, assume

$$\sum_{r=1}^{k} \frac{1}{r(r+1)(r+2)} = \frac{k(k+3)}{4(k+1)(k+2)}$$

Inductive step: Starting with the left-hand side for $n = k + 1$:

$$\begin{aligned}
\sum_{r=1}^{k+1} \frac{1}{r(r+1)(r+2)} &= \sum_{r=1}^{k} \frac{1}{r(r+1)(r+2)} + \frac{1}{(k+1)(k+2)(k+3)} \\
&= \frac{k(k+3)}{4(k+1)(k+2)} + \frac{1}{(k+1)(k+2)(k+3)} \qquad \text{(by the inductive hypothesis)} \\
&= \frac{k(k+3)(k+3)}{4(k+1)(k+2)(k+3)} + \frac{4}{4(k+1)(k+2)(k+3)} \\
&= \frac{k(k+3)^2 + 4}{4(k+1)(k+2)(k+3)} \\
&= \frac{k(k^2+6k+9) + 4}{4(k+1)(k+2)(k+3)} \\
&= \frac{k^3 + 6k^2 + 9k + 4}{4(k+1)(k+2)(k+3)} \\
&= \frac{(k+1)^2(k+4)}{4(k+1)(k+2)(k+3)} \\
&= \frac{(k+1)(k+4)}{4(k+2)(k+3)}
\end{aligned}$$

This is the right-hand side with $n = k + 1$, since

$$\frac{(k+1)(k+4)}{4(k+2)(k+3)} = \frac{(k+1)\big((k+1)+3\big)}{4\big((k+1)+1\big)\big((k+1)+2\big)}$$

so the statement is true for $n = k + 1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k + 1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all $n \in \mathbb{Z}^+$.

9. **(a)** Prove by induction that for all positive integers n

$$\sum_{r=1}^{n}(2r-1)^3 = n^2(2n^2-1)$$

[6]

(b) Given that

$$\sum_{r=1}^{2n}(2r-1)^3 = k\sum_{r=1}^{n}(2r-1)^3$$

show that

$$n^2 = \frac{k-4}{2(k-16)}$$

[5]

Solution

(a) Let

$$S_n = \sum_{r=1}^{n}(2r-1)^3$$

We prove by induction that

$$S_n = n^2(2n^2-1)$$

for all positive integers n.

Base case: $n = 1$. The left-hand side is

$$S_1 = (2\times 1-1)^3 = 1^3 = 1$$

and the right-hand side is

$$1^2(2\times 1^2-1) = 1(2-1) = 1$$

so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$S_k = \sum_{r=1}^{k}(2r-1)^3 = k^2(2k^2-1)$$

Inductive step: Starting from S_{k+1}:

$$\begin{aligned} S_{k+1} &= \sum_{r=1}^{k+1}(2r-1)^3 \\ &= \sum_{r=1}^{k}(2r-1)^3 + (2(k+1)-1)^3 \\ &= S_k + (2k+1)^3 \end{aligned}$$

Now use the inductive hypothesis:

$$\begin{aligned} S_{k+1} &= k^2(2k^2-1)+(2k+1)^3 \\ &= 2k^4-k^2+(8k^3+12k^2+6k+1) \\ &= 2k^4+8k^3+11k^2+6k+1 \end{aligned}$$

We now factor this in the required form:

$$\begin{aligned} (k+1)^2\big(2(k+1)^2-1\big) &= (k+1)^2(2k^2+4k+1) \\ &= (k^2+2k+1)(2k^2+4k+1) \\ &= 2k^4+8k^3+11k^2+6k+1 \end{aligned}$$

Hence

$$S_{k+1} = (k+1)^2\big(2(k+1)^2-1\big)$$

so the statement is true for $n = k+1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all positive integers n.

(b) From part (a):

$$\sum_{r=1}^{n}(2r-1)^3 = n^2(2n^2-1)$$

With n replaced by $2n$:

$$\begin{aligned} \sum_{r=1}^{2n}(2r-1)^3 &= (2n)^2\big(2(2n)^2-1\big) \\ &= 4n^2(8n^2-1) \end{aligned}$$

We are given that

$$\sum_{r=1}^{2n}(2r-1)^3 = k\sum_{r=1}^{n}(2r-1)^3$$

Substitute the formula from part (a):

$$4n^2(8n^2-1) = k\big(n^2(2n^2-1)\big)$$

Since n is a positive integer, $n^2 \neq 0$, so divide through by n^2:

$$4(8n^2-1) = k(2n^2-1)$$

Expand:

$$32n^2-4 = 2kn^2-k$$

Rearranging:

$$\begin{aligned} 32n^2-2kn^2 &= 4-k \\ (32-2k)n^2 &= 4-k \end{aligned}$$

Multiply both sides by -1:

$$(2k-32)n^2 = k-4$$

Hence

$$n^2 = \frac{k-4}{2k-32} = \frac{k-4}{2(k-16)}$$

10. Prove by induction that for all positive integers n

$$\sum_{r=1}^{n} \log\left(1 - \frac{1}{4r^2}\right) = \log\left(\frac{(2n)!(2n+1)!}{2^{4n}(n!)^4}\right) \quad \textbf{[6]}$$

Solution

Let

$$S_n = \sum_{r=1}^{n} \log\left(1 - \frac{1}{4r^2}\right)$$

We prove by induction that

$$S_n = \log\left(\frac{(2n)!(2n+1)!}{2^{4n}(n!)^4}\right)$$

for all positive integers n.

Base case: $n = 1$. The left-hand side is

$$S_1 = \log\left(1 - \frac{1}{4}\right) = \log\left(\frac{3}{4}\right)$$

and the right-hand side is

$$\log\left(\frac{(2\times 1)!(2\times 1+1)!}{2^{4\times 1}(1!)^4}\right) = \log\left(\frac{2!\,3!}{2^4}\right) = \log\left(\frac{2\times 6}{16}\right) = \log\left(\frac{3}{4}\right)$$

so the statement is true for $n = 1$.

Inductive hypothesis: Assume the statement is true for $n = k$, where k is a positive integer. That is, assume

$$S_k = \log\left(\frac{(2k)!(2k+1)!}{2^{4k}(k!)^4}\right)$$

Inductive step: We must show that

$$S_{k+1} = \log\left(\frac{(2k+2)!(2k+3)!}{2^{4k+4}((k+1)!)^4}\right)$$

Starting from the definition of S_{k+1}:

$$S_{k+1} = S_k + \log\left(1 - \frac{1}{4(k+1)^2}\right)$$

First simplify the extra term:

$$\begin{aligned} 1 - \frac{1}{4(k+1)^2} &= \frac{4(k+1)^2 - 1}{4(k+1)^2} \\ &= \frac{(2k+2)^2 - 1}{(2k+2)^2} \\ &= \frac{(2k+1)(2k+3)}{(2k+2)^2} \end{aligned}$$

So, using the inductive hypothesis and the law $\log a + \log b = \log(ab)$:

$$S_{k+1} = \log\left(\frac{(2k)!(2k+1)!}{2^{4k}(k!)^4}\right) + \log\left(\frac{(2k+1)(2k+3)}{(2k+2)^2}\right)$$
$$= \log\left(\frac{(2k)!(2k+1)!(2k+1)(2k+3)}{2^{4k}(k!)^4(2k+2)^2}\right)$$

Now use the factorial identities:

$$(2k+2)! = (2k+2)(2k+1)(2k)!$$

and

$$(2k+3)! = (2k+3)(2k+2)(2k+1)!$$

Multiplying these together gives

$$(2k+2)!(2k+3)! = (2k+2)(2k+1)(2k)!\,(2k+3)(2k+2)(2k+1)!$$
$$= (2k+2)^2(2k+1)(2k+3)(2k)!(2k+1)!$$

Therefore

$$(2k)!(2k+1)!(2k+1)(2k+3) = \frac{(2k+2)!(2k+3)!}{(2k+2)^2}$$

Substituting this into the expression for S_{k+1}:

$$S_{k+1} = \log\left(\frac{(2k+2)!(2k+3)!}{2^{4k}(k!)^4(2k+2)^4}\right)$$

Now simplify the denominator:

$$(2k+2)^4 = [2(k+1)]^4 = 2^4(k+1)^4$$

and

$$((k+1)!)^4 = ((k+1)k!)^4 = (k+1)^4(k!)^4$$

Hence

$$2^{4k}(k!)^4(2k+2)^4 = 2^{4k} \times 2^4 \times (k+1)^4(k!)^4 = 2^{4k+4}((k+1)!)^4$$

So

$$S_{k+1} = \log\left(\frac{(2k+2)!(2k+3)!}{2^{4k+4}((k+1)!)^4}\right)$$

This is the right-hand side with $n = k+1$, so the statement is true for $n = k+1$.

Conclusion: If the statement is true for $n = k$, then it is true for $n = k+1$. Since it is true for $n = 1$, by the principle of mathematical induction the statement is true for all positive integers n.